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Interpreting user movement as direct touch user interface interactions

Abstract

Various implementations disclosed interpret direct touch-based gestures, such as drag and scroll gestures, made by a user virtually touching one position of a user interface and moving their hand to another position of the user interface. For example, such gestures may be made relative to a user interface presented in an extended reality (XR) environment. In some implementations, a user movement is interpreted using one or more techniques that avoid unexpected gain or loss of user-interface-associated motion. Some implementations determine which segments of a movement to associate with user interface content based on characteristics of the movement. Some implementations determine that a break occurs when a user movement leaves a break volume that is adjusted dynamically.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/409,326 filed Sep. 23, 2022, which is incorporated herein in its entirety.

TECHNICAL FIELD
(1) The present disclosure generally relates to assessing user interactions with electronic devices that involve hand and body movements.
BACKGROUND
(2) Existing user interaction systems may be improved with respect to facilitating interactions based on user hand and body movements and other activities.
SUMMARY
(3) Various implementations disclosed herein include devices, systems, and methods that interpret direct touch-based gestures, such as drag and swipe gestures, made by a user virtually touching one position of a user interface and, while still touching, moving their hand to another position of the user interface (UI). Such gestures may be made relative to a user interface presented as virtual content in the 3D space of an extended reality (XR) environment. Ideally such gestures would be associated with user interface positions based on where the user's hand virtually intersects the user interface, e.g., where the hand makes contact and breaks contact with the user interface. However, because a user's perception of when and where the user is virtually touching the user interface (e.g., overlapping the user interface in an extended reality (XR) space) may be inaccurate, unexpected gain or loss of user interface-associated motion (referred to as “hooking”) may occur. For example, a segment of the user's movement may be associated with user interface contact when the user expects the segment of movement to not occur during user interface contact. Conversely, a segment of the user's movement may not be associated with user interface contact when the user expects the segment of movement to occur during user interface contact.
(4) Some implementations determine which segments of a movement to associate with user interface content based on characteristics of the movement. In drags (i.e., where a user attempts to touch at a position on the user interface move to a second position on the user interface and release the touch at that second position), hooking can occur when a segment of the movement associated with retracting the hand is associated with UI contact, in contrast to the user's expectation that such retracting would not occur during UI contact. This may cause the system to identify an incorrect break point on the user interface, i.e., using the retraction portion of the movement to identify the break point rather than the position on the user interface corresponding to the user's position when the intentional UI-contacting motion ceased. Some implementations avoid such erroneous associations (and thus more accurately interpret movements) by determining whether to associate such a segment (e.g., a potential retraction segment) based on whether the characteristics of the segment are indicative of a retraction. In other words, some implementations determine that a segment of a movement that would otherwise be associated with user interface content (e.g., based

on actual position overlap) should not associated be associated with user interface contact if the segment of the motion is likely to be a retraction. This may involve determining to not associate a segment of motion with user interface contact based on determining that the segment is a likely to be a retraction based on assessing how aligned the segment is with a retraction axis, a significance of a retraction direction change, or a motion stop.

(5) In some implementations, a processor performs a method by executing instructions stored on a computer readable medium. The method displays an XR environment corresponding to a 3D environment, where the XR environment comprises a user interface and a movement (e.g., of a user's finger or hand). The method determines whether each of multiple segments of the movement has a characteristic that satisfies a retraction criterion. The retraction criterion is configured to distinguish retraction motion from another type of motion. As examples, the characteristic may be, but is not limited to being, (a) a measure of alignment between the movement direction during the respective segment and a retraction direction (b) a measure how quickly movement direction changes and/or (c) whether the user (e.g., hand/finger) has stopped moving. The method associates a subset of the segments of the movement with user interface contact based on whether the characteristic of each of the segments satisfies the retraction criterion. In some implementations, the association of select segments is achieved by implementing a retraction dead-band such that movement occurring during the retraction (because such movement is within the retraction dead-band) is not recognized as user interface contact motion.

(6) In some implementations, user movement is interpreted using a technique that avoids unexpected gain or loss of UI-associated motion using a dynamic break volume. Some implementations determine that a break occurs when a user movement leaves a break volume that is adjusted dynamically based on retraction confidence and/or piercing depth. Intentional swipe momentum may be preserved by breaking at an appropriate time before motion is lost from an arc or retraction.

(7) In some implementations, a processor performs a method by executing instructions stored on a computer readable medium. The method displays an XR environment corresponding to a 3D environment, where the XR environment comprises a user interface and a movement. The method adjusts a break volume based on the movement, the break volume defining a region of the XR environment in which the movement will be associated with user interface contact. In some examples, the break volume is positionally shifted based on retraction confidence. In some implementations, a slope or other shape attribute of the break volume is adjusted based on a piercing depth. The method determines to discontinue associating the movement with user interface contact (e.g., determining that a break event has occurred) based on the movement crossing a boundary of the break volume.

(8) In accordance with some implementations, a device includes one or more processors, a non-transitory memory, and one or more programs; the one or more programs are stored in the non-transitory memory and configured to be executed by the one or more processors and the one or more programs include instructions for performing or causing performance of any of the methods described herein. In accordance with some implementations, a non-transitory computer readable storage medium has stored therein instructions, which, when executed by one or more processors of a device, cause the device to perform or cause performance of any of the methods described herein. In accordance with some implementations, a device includes: one or more processors, a non-transitory memory, and means for performing or causing performance of any of the methods described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the present disclosure can be understood by those of ordinary skill in the art, a more detailed description may be had by reference to aspects of some illustrative implementations, some of which are shown in the accompanying drawings.

(2) FIG. **1** illustrates an exemplary electronic device operating in a physical environment in accordance with some implementations.

(3) FIG. **2** illustrates views of an XR environment provided by the device of FIG. **1** based on the physical environment of FIG. **1** in which a movement including an intentional drag is interpreted, in accordance with some implementations.

(4) FIG. **3** illustrates interpreting a user's intentions in making a movement relative to an actual user interface position.

(5) FIG. **4** illustrates interpreting a user's intentions in making a movement relative to an actual user interface position.

(6) FIGS. **5-6** illustrate a movement having characteristics corresponding to a retraction in accordance with some implementations.

(7) FIG. **7** illustrates a retraction dead-band in accordance with some implementations.

(8) FIGS. **8-9** illustrate a dynamic break volume in accordance with some implementations.

(9) FIGS. **10-11** illustrate a trajectory correction in accordance with some implementations.

(10) FIG. **12** is a flowchart illustrating a method for determining which segments of a movement to associate with user interface content based on characteristics of the movement, in accordance with some implementations.

(11) FIG. **13** is a flowchart illustrating a method for interpreting a movement using a dynamic break volume in accordance with some implementations.

(12) FIG. **14** is a block diagram of an electronic device of in accordance with some implementations.

(13) In accordance with common practice the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

DESCRIPTION

(14) Numerous details are described in order to provide a thorough understanding of the example implementations shown in the drawings. However, the drawings merely show some example aspects of the present disclosure and are therefore not to be considered limiting. Those of ordinary skill in the art will appreciate that other effective aspects and/or variants do not include all of the specific details described herein. Moreover, well-known systems, methods, components, devices and circuits have not been described in exhaustive detail so as not to obscure more pertinent aspects of the example implementations described herein.

(15) FIG. **1** illustrates an exemplary electronic device **110** operating in a physical environment **100**. In this example of FIG. **1**, the physical environment **100** is a room that includes a desk **120**. The electronic device **110** includes one or more cameras, microphones, depth sensors, or other sensors that can be used to capture information about and evaluate the physical environment **100** and the objects within it, as well as information about the user **102** of the electronic device **110**. The information about the physical environment **100** and/or user **102** may be used to provide visual and audio content and/or to identify the current location of the physical environment **100** and/or the location of the user within the physical environment **100**.

(16) In some implementations, views of an extended reality (XR) environment may be provided to one or more participants (e.g., user **102** and/or other participants not shown). Such an XR environment may include views of a 3D environment that is generated based on camera images

and/or depth camera images of the physical environment **100** as well as a representation of user **102** based on camera images and/or depth camera images of the user **102**. Such an XR environment may include virtual content that is positioned at 3D locations relative to a 3D coordinate system associated with the XR environment, which may correspond to a 3D coordinate system of the physical environment **100**.

(17) People may sense or interact with a physical environment or world without using an electronic device. Physical features, such as a physical object or surface, may be included within a physical environment. For instance, a physical environment may correspond to a physical city having physical buildings, roads, and vehicles. People may directly sense or interact with a physical environment through various means, such as smell, sight, taste, hearing, and touch. This can be in contrast to an extended reality (XR) environment that may refer to a partially or wholly simulated environment that people may sense or interact with using an electronic device. The XR environment may include virtual reality (VR) content, mixed reality (MR) content, augmented reality (AR) content, or the like. Using an XR system, a portion of a person's physical motions, or representations thereof, may be tracked and, in response, properties of virtual objects in the XR environment may be changed in a way that complies with at least one law of nature. For example, the XR system may detect a user's head movement and adjust auditory and graphical content presented to the user in a way that simulates how sounds and views would change in a physical environment. In other examples, the XR system may detect movement of an electronic device (e.g., a laptop, tablet, mobile phone, or the like) presenting the XR environment. Accordingly, the XR system may adjust auditory and graphical content presented to the user in a way that simulates how sounds and views would change in a physical environment. In some instances, other inputs, such as a representation of physical motion (e.g., a voice command), may cause the XR system to adjust properties of graphical content.

(18) Numerous types of electronic systems may allow a user to sense or interact with an XR environment. A non-exhaustive list of examples includes lenses having integrated display capability to be placed on a user's eyes (e.g., contact lenses), heads-up displays (HUDs), projection-based systems, head mountable systems, windows or windshields having integrated display technology, headphones/earphones, input systems with or without haptic feedback (e.g., handheld or wearable controllers), smartphones, tablets, desktop/laptop computers, and speaker arrays. Head mountable systems may include an opaque display and one or more speakers. Other head mountable systems may be configured to receive an opaque external display, such as that of a smartphone. Head mountable systems may capture images/video of the physical environment using one or more image sensors or capture audio of the physical environment using one or more microphones. Instead of an opaque display, some head mountable systems may include a transparent or translucent display. Transparent or translucent displays may direct light representative of images to a user's eyes through a medium, such as a hologram medium, optical waveguide, an optical combiner, optical reflector, other similar technologies, or combinations thereof. Various display technologies, such as liquid crystal on silicon, LEDs, uLEDs, OLEDs, laser scanning light source, digital light projection, or combinations thereof, may be used. In some examples, the transparent or translucent display may be selectively controlled to become opaque. Projection-based systems may utilize retinal projection technology that projects images onto a user's retina or may project virtual content into the physical environment, such as onto a physical surface or as a hologram.

(19) FIG. **2** illustrates views **210***a-e* of an XR environment provided by the device of FIG. **1** based on the physical environment of FIG. **1** in which a user movement is interpreted. The views **210***a-e* of the XR environment include an exemplary user interface **230** of an application (i.e., virtual content) and a depiction **220** of the table **120** (i.e., real content). Providing such a view may involve determining 3D attributes of the physical environment **100** and positioning the virtual content, e.g., user interface **230**, in a 3D coordinate system corresponding to that physical environment **100**.

(20) In the example of FIG. **2**, the user interface **230** may include various content and user interface elements, including a scroll bar shaft **240** and its scroll bar handle **242** (also known as a scroll bar thumb). Interactions with the scroll bar handle **242** may be used by the user **202** to provide input to which the user interface **230** respond, e.g., by scrolling displayed content or otherwise. The user interface **230** may be flat (e.g., planar or curved planar without depth). Displaying the user interface **230** as a flat surface may provide various advantages. Doing so may provide an easy to understand or otherwise use portion of an XR environment for accessing the user interface of the application.
(21) The user interface **230** may be a user interface of an application, as illustrated in this example. The user interface **230** is simplified for purposes of illustration and user interfaces in practice may include any degree of complexity, any number of user interface elements, and/or combinations of 2D and/or 3D content. The user interface **230** may be provided by operating systems and/or applications of various types including, but not limited to, messaging applications, web browser applications, content viewing applications, content creation and editing applications, or any other applications that can display, present, or otherwise use visual and/or audio content.
(22) In some implementations, multiple user interfaces (e.g., corresponding to multiple, different applications) are presented sequentially and/or simultaneously within an XR environment using one or more flat background portions. In some implementations, the positions and/or orientations of such one or more user interfaces may be determined to facilitate visibility and/or use. The one or more user interfaces may be at fixed positions and orientations within the 3D environment. In such cases, user movements (e.g., of a user moving their head while wearing an HMD) would not affect the position or orientation of the user interfaces within the 3D environment.
(23) In other implementations, the one or more user interfaces may be body-locked content, e.g., having a distance and orientation offset relative to a portion of the user's body (e.g., their torso). For example, the body-locked content of a user interface could be 2 meters away and 45 degrees to the left of the user's torso's forward-facing vector. While wearing an HMD, if the user's head turns while the torso remains static, a body-locked user interface would appear to remain stationary in the 3D environment at 2 m away and 45 degrees to the left of the torso's front facing vector. However, if the user does rotate their torso (e.g., by spinning around in their chair), the body-locked user interface would follow the torso rotation and be repositioned within the 3D environment such that it is still 2 m away and 45 degrees to the left of their torso's new forward-facing vector.
(24) In other implementations, user interface content is defined at a specific distance from the user with the orientation relative to the user remaining static (e.g., if initially displayed in a cardinal direction, it will remain in that cardinal direction regardless of any head or body movement). In this example, the orientation of the body-locked content would not be referenced to any part of the user's body. In this different implementation, the body-locked user interface would not reposition itself in accordance with the torso rotation. For example, body-locked user interface may be defined to be 2 m away and, based on the direction the user is currently facing, may be initially displayed north of the user. If the user rotates their torso 180 degrees to face south, the body-locked user interface would remain 2 m away to the north of the user, which is now directly behind the user.
(25) A body-locked user interface could also be configured to always remain gravity or horizon aligned, such that head and/or body changes in the roll orientation would not cause the body-locked user interface to move within the 3D environment. Translational movement would cause the body-locked content to be repositioned within the 3D environment in order to maintain the distance offset.
(26) In the example of FIG. **2**, at a first instant in time corresponding to view **210***a*, the user **102** has positioned their hand in the physical environment **100** and a corresponding depiction **202** of the user **102** shows a fingertip of the user **102** not yet touching the user interface **230**. The device **110**

may track user positioning, e.g., locations of the user's fingers, hands, arms, etc.
(27) The user **102** moves their hand/finger forward in the physical environment **100** causing a corresponding movement of the depiction **202** of the user **102**. Thus, at a second instant in time corresponding to the view **210***b*, the user **102** has positioned their hand in the physical environment **100** and a corresponding depiction **202** of the user **102** shows a fingertip of the user **102** touching or extending into a scroll bar handle **242**.
(28) The device **110** may determine positioning of the user relative to the user interface **230** (e.g., within an XR environment) and identify user interactions with the user interface based on the positional relationships between them and/or information indicative of when the user is perceiving or expecting their hand/finger to be in contact with the user interface. In this example, the device **110** detects a make point (e.g., a point in time and/or the 3D space at which contact between a user and a user interface occurs or is expected to occur) as the portion of the depiction **202** of the fingertip of the user **102** contacts the scroll bar handle **242**.
(29) Detecting such a make point may initiate a user interaction. For example, the device **110** may start tracking subsequent movement corresponding to a drag type user interaction that will be interpreted to move the scroll bar handle **242** along or otherwise based on the right/left movement of the depiction **202** of the portion of the user **102**. Movement of the scroll bar handle **242** (caused by such user motion) may also trigger a corresponding user interface response, e.g., causing the user interface **230** to scroll displayed content according to the amount the scroll bar handle **242** is moved, etc.
(30) In the example of FIG. **2**, at a third instant in time corresponding to view **210***c*, the user **102** has moved their hand in the physical environment **100** and a corresponding depiction **202** of the user **102** has moved left with respect to the user interface **230** while the hand is still considered to be in contact with the user interface **230**. Movement of the hand may continue to drag the scroll bar handle **242** in this way until a break point (e.g., a point in time and/or the 3D space at which contact between a user and a user interface occurs or is expected to be discontinued).
(31) In this example, at a fourth instant in time corresponding to view **210***d*, the user **102** has continued moving their hand in the physical environment **100** and a corresponding depiction **202** of the user **102** has continued moving left with respect to the user interface **230** since the hand is still considered to be in contact with the user interface until it reaches break point **250**. At the fifth instant in time corresponding to view **210***e*, the device **110** detects that the user has concluding the drag type user interaction and the hand is retracting as shown by the depiction **202**. The segment of the user movement (e.g., movement after break point **250** at which the user begins retracting the depiction **202** away from the user interface **230**) is not associated with user interface contact, e.g., it is not interpreted as part of the drag-type user interaction.
(32) Implementations disclosed herein interpret user movements that relate to the positioning of a user interface within a 3D space so that the user movements are interpreted as direct touches with the user interface in accordance with user expectations, e.g., when the user perceives or thinks they are virtually contacting the user interface, which may not necessarily correlate precisely with when actual contact occurs between the user and the user interface depictions in the XR environment.
(33) Some implementations determine which segments of a movement to associate with user interface content based on characteristics of the movement. In drags (i.e., where a user attempts to touch at a position on the user interface move to a second position on the user interface and release the touch at that second position), hooking can occur when a segment of the movement associated with retracting the hand is associated with UI contact in contrast to the user's expectation that such retracting would not occur during UI contact. This may cause the system to identify an incorrect break point on the user interface, i.e., using the retraction to identify the break point rather than the position on the user interface corresponding to the user's position when the drag motion ceased.
(34) Some implementations avoid such erroneous associations (and thus more accurately interpret movements) by determining whether to associate such a segment (e.g., a potential retraction

segment) based on whether the characteristics of the segment are indicative of a retraction. In other words, some implementations determine that a segment of a movement that would otherwise be associated with user interface content (e.g., based on actual position overlap) should not associated be associated with user interface contact if the segment of the motion is a retraction. This may involve determining to not associate a segment of motion with user interface contact based on determining that the segment is a retraction based on (a) assessing how aligned the segment is with a retraction axis, (b) a significance of a retraction direction change, or (c) a motion stop.
(35) FIG. **3** illustrates a user's intentions in making a movement relative to an actual user interface position. In this example, during a first segment **301** of a user movement, the user **310** moves a portion of their body (e.g., their finger, hand, etc.) with the intention of making contact with a user interface. In this example, the first segment **301** of the movement extends through the actual UI plane **305** to perceived UI plane **304**. The user may perceive (or otherwise expect) that the UI plane is at a location that differs from its actual position for various reasons.
(36) Based on the user's perception of where the UI plane is, i.e., perceived UI plane **304** location, the user continues moving the portion of their body (e.g., their finger, hand, etc.) during a second segment **302** of movement in a drag-type motion, e.g., moving their finger across the user interface. The actual motion path during such a second segment **302** may be linear or non-linear (e.g., arcuate as illustrated). In this example, based on the movement during the first segment **301** and/or the second segment **302**, the device **110** determines a location of a make point **315** on the actual user interface **305**. In one example, the change in direction exceeding a threshold is determined as the time of the make point **315** and the make point **315** location is determined based on where the movement intersected the actual UI plane **305**. In another example, the position **306** at which such a change occurred is used to determine a corresponding position on the actual UI plane **305** to use as the make point.
(37) After the make point is established, the movement of the user is monitored and used as user input. The movement is used as input (i.e., continues to be associated with contact with the user interface) until a condition is satisfied, e.g., a break point is determined.
(38) In this example, based on the user's perception of where the UI plane is, i.e., perceived UI plane **304** location, at the end of the intended drag motion which occurs at the end of the second segment **302**, the user moves the portion of their body (e.g., their finger, hand, etc.) during a third segment **303** of movement in a retraction movement back towards themselves. During the second segment **302** and the third segment **303** of the movement, the movement is assessed to attempt to identify when and where the user expects that UI contact has concluded. This assessment may occur repeatedly, e.g., every frame, every 5 frames, every 0.1 ms, etc.) such that the association of the movement with user interface contact can be determined as soon as (or very soon after) the user stops intending to make contact with the user interface. This may involve assessing the path of the movement to determine whether a current segment of the movement has a characteristic that satisfies a retraction criterion. Such a retraction criterion may be configured to distinguish retraction motion from another type of motion (e.g., continued drag motion, swiping motion, etc.). The characteristic may be, but is not limited to being, (a) a measure of alignment between the movement direction and a retraction direction (b) a measure of retraction direction change and/or (c) whether the user (e.g., finger) has stopped.
(39) In the example of FIG. **3**, the third segment **303** is determined to be a retraction motion. Accordingly, this third segment **303** is not treated as movement associated with user interface contact/drag input. Only the second segment **302** is treated as movement associated with user interface contact/drag input. The assessment of whether segments should be associated with user interface contact or not may be used to determine an appropriate break point for the movement. In this example, the second **302** segment transitions at point **307** to the third segment **303**, i.e., association of the movement with user interface contact is determined to end at this point in time. This is used to determine a corresponding position **330** on the actual user interface **305** to use as the

break point rather than the position **320** at which the user's retracting body portion (e.g., hand, finger, etc.) crossed the actual user interface **305**. FIGS. **5-7**, described below, provide additional examples of using movement characteristics to interpret segments of user movement, e.g., with respect to determine which segments should be associated with user interface contact.
(40) FIG. **4** also illustrates a user's intentions in making a movement relative to an actual user interface position. In this example, the user **410** makes a swiping movement of the portion of their body (e.g., their finger, hand, etc.). In this example, the first segment **401** of the movement swipes through the actual UI plane **405** into perceived UI plane **404**. Based on the user's perception of where the UI plane is, i.e., perceived UI plane **404** location, the user continues making the swiping movement during a second segment **402** and through a third segment **403** during which the swiping motion broadly arcs back towards the user. The end of the swipe may differ from a drag retraction (e.g., as illustrated in FIG. **3**) and in the movement may be used to identify the type of movement (e.g., drag or swipe) and/or treat the end of the movements (e.g., third segments **303**, **403**) differently.
(41) In some implementations, the swiping movement illustrated in FIG. **4** is interpreting using a dynamic break volume to avoid unexpected gain or loss of UI-associated motion. This may involve determining that a break event occurs based on determining that the movement leaves a break volume that is adjusted dynamically based on (a) retraction confidence and/or (b) piercing depth. Intentional swipe momentum may be preserved by breaking at an appropriate time before motion is lost from an arc or retraction for example using swipe trajectory correction. FIGS. **8-11**, described below, provide additional examples of using dynamic break volumes and correcting trajectory (e.g., swipe trajectory).
(42) FIGS. **5-6** illustrate a segment of a movement having characteristics corresponding to a drag motion followed by a retraction motion. In this example, the user movement (e.g. of user **510**) includes a drag segment **502** and a retraction segment **503** relative to the actual user interface **505**. The movement transitions from the drag segment **502** to the retraction segment **503** at point **503**. This transition is detected based on detecting that the retraction segment **503** has one or more characteristics that correspond to a retraction. In this example, a retraction direction **510** is identified based on the current position of the user **510** (e.g., finger, hand, etc.) and the user's head **520**. In other examples, a retraction direction may be based on another portion of the user, e.g., the direction between the current position of the user **510** (e.g., finger, hand, etc.) and a center of the user's torso (not shown).
(43) The retraction direction **510** may be used to determine a retraction confidence, e.g., a measure of confidence that a current segment of the movement corresponds to a retraction versus another type of motion. For example, such a retraction confidence may be based on how aligned the segment is with the retraction motion. Movement that is more aligned with the retraction direction **510** may be more likely to correspond to drag retraction movement than movement that is not aligned with (e.g., perpendicular to, etc.) the retraction direction **510**. In this example, the retraction segment **503** of the movement is closely aligned with the retraction direction **510** and thus the segment is determined to be a retraction following the drag.
(44) In some implementations, movement characteristics are used to detect retraction and/or trigger determining an early break event (i.e., prior to the user actually disconnecting from the user interface).
(45) In some implementations, rather than using an instantaneous movement direction (e.g., direction **603**) to compare with a retraction direction **515** to identify retractions, an averaged movement direction (**604**) may be determined and compared with a retraction direction **515** to identify retractions. This may help ensure that noise or micro-changes of direction do not inadvertently trigger a retraction detection. For example, it may be more accurate to use an averaged movement direction **604** than a current instantaneous movement direction **603** to identify retractions.

(46) In some implementations, an average movement direction (e.g., movement **604**) is determined using a lag position **504** (e.g., an index finger tip lag position) and used to assess a retraction confidence. Such a lag position **504** may be a lazy follow of the user's position (e.g., finger position) determined using a delayed moving average filter (50 ms, 125 ms). The lag position **504** may be used to determine an average movement direction (î) **604** from that lag position **504** to the current position **508**, e.g., î=norm(current finger position−lag position). A retraction axis/direction (ř) **510**, e.g., ř=norm(headpos−current finger position). The current movement direction (î) **604** and the retraction axis/direction (ř) **515** may be used to determine a retraction confidence based on their dot product: r.sub.c=î.Math.ř. In this example, a r.sub.c=1.0 is indicative of a highly confident retraction, a r.sub.c=−1.0 is indicative of a highly confident piercing type movement, and a r.sub.c=0.0 is indicative of a low confidence retraction (not retracting). Retraction confidence may be overridden or automatically set to zero in circumstances in which sensor data providing trajectory information is uncertain or otherwise when the trajectory of the movement is not trusted.
(47) FIG. **7** illustrates a retraction dead-band **720**. Following the example, of FIGS. **5-6**, a retraction dead-band **720** is spawned based on detecting the occurrence of motion corresponding to a retraction. The retraction dead-band **720** is a region or volume of 3D space used to interpret movement, e.g., hand movement within the retraction dead-band **720** is considered a retraction. However, if the user motion leaves the retraction dead-band **720** 3D space, it may no longer be considered a retraction and thus may be interpreted as input, e.g., recognized as a tap, drag, swipe, etc. A retraction dead-band may be used to distinguish motion corresponding to an input versus a movement corresponding to a retraction. The retraction dead-band may be shaped, positioned, and otherwise configured so that movement closer to the user interface **505** will be more likely to be outside of the retraction dead-band **720** than movement further from the user interface **505**, and thus more likely to be interpreted as a continuous scroll, drag, etc. The retraction dead-band **720** may have various shapes, e.g., having a straight profile or a curved (e.g., exponentially curved) profile.
(48) In FIG. **7**, the retraction dead-band **720** is aligned with (e.g., centered on) the retraction axis/direction **515** such that any in-plane motion is discarded. Movement during the retraction segment **503** that is within the retraction dead-band **720** will not be associated with user interface contact, e.g., will not continue to affect the drag response. However, if the movement exits the retraction dead-band **720**, it may resume being treated as movement associated with user interface contact. The retraction dead-band **720** may be configured to timeout after a threshold amount of time.
(49) FIGS. **8-9** illustrate a dynamic break volume. Such a break volume may be particularly useful with respect to swipe type input. Swipes tend to be faster than drag interactions and have more arc. When swiping, a user may expect to preserve the motion/velocity at the point in time/space when the perceive that UI contact is broken. For example, the user may swipe and expect the swipe to initiate a scroll that continues after UI contact is broken based on the speed of movement when the UI content ends. However, this perceived break may not coincide precisely with the actual break of contact from the user interface. Some implementations disclosed herein utilize a dynamic break volume to, among other things, preserve the user's intentional swipe momentum, e.g., by breaking early before motion is lost from an arc or retraction.
(50) FIG. **8** illustrates a user movement **802** (of user **810**) relative to a user interface **805**. A break volume **815** is generated and used to determine when to break the swipe motion, i.e., discontinue associating the movement **802** with user interface contact. The break volume **815** may be adjusted in shape or position over time, for example, based on the current position of the user **810** or a position (e.g., a lag position) determined based on the current position of the user **810**. In this example, an axis **830** of the break volume **815** is aligned with a target axis (e.g., the z axis of a user interface **805** based on a current lag position **812**). The current lag position **812** may be determined based on the current user position **813**, e.g., based on lag parameters e.g., a predetermined lag

period, lag distance, etc. In this example, the break volume **815** is a centroid C.sub.xy that tracks a lag (e.g., index.sub.lag **820** associated with an index finger tip position). The break volume **815** may be configured to change shape, position, and/or orientation based the movement **802** and/or during the movement **802**. The break volume **815** may expand and contract in an umbrella-like way remaining symmetrical about the axis **830** while potentially shifting laterally relative to the user interface (e.g., shifting down in FIG. **8**. The break volume **815** may be shifted based on retraction confidence, and/or be increased in slope based on piercing direction depth **825** (e.g., tracking index.sub.lag **820**).

(51) In some implementations, a break volume **815** is not symmetrical, e.g., not symmetrical about axis **830**. For example, a break volume **815** may include only a lower portion below the axis **830**. In some implementations, a break volume **815** is symmetrical about an axis that is not perpendicular/orthogonal to user interface **805**. For example, a break volume may be symmetrical about an axis that is at a predetermined angle relative the user interface, where the predetermined angle is determined based on user-specific characteristics, e.g., the user's typical motion path characteristics when making a gesture of a given type.

(52) In an alternative implementation, break volume **815** is determined based on a predicted path, e.g., based trajectory, speed, or other characteristics of a user motion. For example, the break volume **815** may be determined based on a predicted path that is predicted when a gesture is initially recognized, e.g., as a swipe gesture, and associated with speed, direction, path or other motion characteristics. In some implementations, based on one or more points along a predicted path, a break volume **815** may be configured with respect to shape and position. In some implementations, a break volume is determined and/or adjusted over time during the course of a user motion based on both a current user position and a predicted user path.

(53) FIG. **9** illustrates a different user movement **902** (of user **910**) relative to a user interface **905**. A break volume **915** is generated and dynamically altered during the movement **902**. The break volume **915** is used to determine when to break the swipe motion, i.e., discontinue associating the movement **902** with user interface contact. In this example, an axis **930** of the break volume **915** is aligned with a target axis (e.g., the z axis of a user interface **905** based on a current lag position). In this example, the break volume **915** is a centroid C.sub.xy that tracks a lag (e.g., index.sub.lag **920** associated with an index finger tip position). The break volume **915** may be configured to change shape, position, and/or orientation based the movement **902** and/or during the movement **902**. The break volume **915** may expand and contract in an umbrella-like way, shifting based on retraction confidence and/or increasing in slope based on piercing direction depth **925** (e.g., tracking index.sub.lag **920**).

(54) FIGS. **8** and **9** illustrate how different movements **802**, **902** can be interpreted using different dynamic break volumes **815**, **915**. Based on the different movements **802**, **902** illustrated in FIGS. **8** and **9**, the respective dynamic break volumes **815**, **915** have different shapes, sizes, and positions. Moreover, during a given movement, the location, shape, and/or orientation of a given break volume is dynamically adjusted to correspond to the current state of the movement. The position of the break volume moves to adapt to the user's current position, depth, and movement path. Using dynamic (context-specific) break volumes may enable a device to better determine break events in different circumstances and ultimately to interpret user movement more consistently with user expectations than when using a fixed (one-size-fits-all break volume).

(55) The shape of the break volumes **815**, **915** may be determined using parameters that allow the break volume to be customized for a particular implementation. Such parameters may include: β (slope sensitivity) corresponding to how sensitive the slope is to changes piercing depth; and a (piercing depth scalar) corresponding to how much the break/volume centroid can shift. These parameters may be used to determine the characteristics of the centroid of the break volumes **815**, **915**. For example, length D.sub.0 **860**, **960** may be determined based on the lag **820**, **920** and the piercing depth scalar: e.g., D.sub.0=index.sub.lag*α. The slope θ **850**, **950** may be determined

based on the length D.sub.0 **860**, **960** and the slope sensitivity: e.g., θ=90−atan 2(D.sub.0,β). The axis C.sub.z **830**, **930** of the break volume **815**, **915** may be determined based on the retraction confidence re (e.g., determined via techniques disclosed herein) and piercing depth **825**, **925**: e.g., C.sub.z=map (|r.sub.c|, depth). The positioning of the break volume **815**, **915** with respect to the other dimensions (e.g., x/y) may depend upon the lag position, e.g., index.sub.lag(xy): e.g., C.sub.xy=index.sub.lag(xy).

(56) FIGS. **10-11** illustrate a trajectory correction based on the movement **802** of FIG. **8**. Natural arcing (e.g., during a swipe) may cause lost motion on break, which may result in UI issues such as "effortful" scrolls. Some implementations preserve intentional swipe velocity on break without introducing noticeable hooks of changes in velocity. Some implementations dampen aggressive hooking that was not broken early via other techniques, e.g., not broken early based on a drag retraction detection.

(57) FIGS. **10-11** illustrate determining a corrected trajectory **1020** to associate with the movement **802** rather than the instantaneous trajectory **1120**. In this example, a lag (i.e., index lag direction (ĥ)) is used to determine the corrected trajectory **1020**. The index lag direction may be determined based on the current index position and a prior index position (e.g., the prior frame's position): e.g., ĥ=norm(index.sub.gt−index.sub.lag). A position difference (Δ pos) may be determined based on the current index position and the prior index position: e.g., Δ pos=index.sub.gt−index.sub.prev. If the segment of the movement has not yet been classified as a drag, the device may predict whether the next frame's Δ (e.g., at position **1103**) will be outside of the break volume **815**. If so, the device makes this frame's positional Δ in line with the direction ĥ, e.g., it corrects the trajectory if the movement is predicted to leave the break volume **815** in the next frame. This technique may suppress some "kick-back" of hooks of failed swipes and should not impact failed drags.

(58) FIG. **12** is a flowchart illustrating a method **1200** for determining which segments of a movement to associate with user interface content based on characteristics of the movement. In some implementations, a device such as electronic device **110** performs method **1200**. In some implementations, method **1200** is performed on a mobile device, desktop, laptop, HMD, or server device. The method **1200** is performed by processing logic, including hardware, firmware, software, or a combination thereof. In some implementations, the method **1200** is performed on a processor executing code stored in a non-transitory computer-readable medium (e.g., a memory).

(59) At block **1202**, the method **1200** displays an XR environment corresponding to a 3D environment, where the XR environment comprises a user interface and a movement. The movement comprising segments.

(60) At block **1204**, the method **1200** determines an occurrence of an event (e.g., a make contact event) associated with contact with the user interface in the XR environment, e.g., based on determining that contact with the UI occurred, was intended to occur, or was perceived by the user. This may involve determining when the user has pierced the user interface. This may involve indicating that a direct touch gesture is in effect, an input criterion (e.g., drag and/or swipe criterion) has been satisfied, and that the movement is being tracked with respect to being input to the user interface.

(61) At block **1206**, the method **1200** determines whether each of the segments of the movement has a characteristic that satisfies a drag retraction criterion. The drag retraction criterion is configured to distinguish retraction motion following a drag from another type of motion. The device may use one or more sensors to track a portion of the user (e.g., the user's hands, finger, finger-tip, index finger-tip, etc.). As examples, the characteristic may be, but is not limited to being, (a) a measure of alignment between the movement direction during the respective segment and a retraction direction (b) a measure how quickly movement direction changes and/or (c) whether the user (e.g., hand/finger) has stopped moving. FIGS. **3**, **5**, and **6** illustrate characteristics that may be used to assess whether a segment satisfies a drag retraction criterion.

(62) In some implementations, the characteristic comprises a drag retraction confidence determined

based on alignment between a direction of the movement during a respective segment and a retraction direction. The retraction direction is a direction from a portion of the user being used for interaction (e.g., finger, hand, etc.) to a head a central portion of the user (e.g., head, torso, etc.). The drag retraction criterion may be whether the drag retraction confidence exceeds a threshold.
(63) In some implementations, the drag retraction criterion is whether a change in the drag retraction confidence exceeds a threshold (e.g., a kink threshold). A rapid change in the drag retraction confidence may correspond to a rapid change in movement direction relative to a retraction axis, which may be indicative that the intended motion of the user touching the user interface has concluded. Similarly, the drag retraction criterion may comprise whether a portion of the user has stopped moving (e.g., is currently moving at a rate below a threshold speed 0.1 m/s). Stopping may be indicative that the intended motion of the user touching the user interface has concluded or that the user has or is about to begin a retraction.
(64) At block **1208**, the method **1200** associates a subset (e.g., one, some, or all) of the segments of the movement with user interface contact based on whether the characteristic of each of the segments satisfies the drag retraction criterion. In some implementations, the association of select segments is achieved by implementing a drag retraction dead-band such that movement occurring during the retraction (because such movement is within the drag retraction dead-band) is not recognized as user interface contact motion. FIG. **7** illustrates an exemplary drag retraction deadband.
(65) FIG. **13** is a flowchart illustrating a method **1300** for interpreting a movement using a dynamic break volume. In some implementations, a device such as electronic device **110** performs method **1300**. In some implementations, method **1300** is performed on a mobile device, desktop, laptop, HMD, or server device. The method **1300** is performed by processing logic, including hardware, firmware, software, or a combination thereof. In some implementations, the method **1300** is performed on a processor executing code stored in a non-transitory computer-readable medium (e.g., a memory).
(66) At block **1302**, the method **1300** displays an XR environment corresponding to a 3D) environment, where the XR environment comprises a user interface and a movement.
(67) At block **1304**, the method **1300** determines an occurrence of an event (e.g., a make contact event) associated with contact with the user interface in the XR environment, e.g., based on determining that contact with the UI occurred, was intended to occur, or was perceived by the user. This may involve determining when the user has pierced the user interface. This may involve indicating that a direct touch gesture is in effect, an input criterion (e.g., drag and/or swipe criterion) has been satisfied, and that the movement is being tracked with respect to being input to the user interface.
(68) At block **1306**, the method **1300** adjusts a break volume based on the movement, the break volume defining a region of the XR environment in which the movement will be associated with user interface contact. Adjusting the break volume may involve shifting the break volume based on a retraction confidence, where the retraction confidence is based on alignment between a direction of the movement and a retraction direction. The retraction direction may be a direction from a portion of the user used for interaction (e.g., hand, finger, etc.) to a central portion of the user (e.g., head, torso, etc.). Adjusting the break volume may involve adjusting a slope of the break volume based on a piercing depth of the movement. Examples of adjusting a break volume are illustrated in FIGS. **8-9**.
(69) At block **1308**, the method **1300** determines to discontinue associating the movement with user interface contact (e.g., determine that a break event has occurred) based on the movement crossing a boundary of the break volume.
(70) In some implementations, a trajectory correction is provided. For example, this may involve adjusting a velocity associated with a first time (e.g., correcting trajectory direction of the current frame) based on determining that the movement will cross outside the boundary of the break

volume at the subsequent time (e.g., next frame). The velocity associated with the first time may be adjusted based on a velocity of a prior time. Examples of trajectory correction are provided in FIGS. **10-11**.

(71) FIG. **14** is a block diagram of electronic device **1400**. Device **1400** illustrates an exemplary device configuration for electronic device **110**. While certain specific features are illustrated, those skilled in the art will appreciate from the present disclosure that various other features have not been illustrated for the sake of brevity, and so as not to obscure more pertinent aspects of the implementations disclosed herein. To that end, as a non-limiting example, in some implementations the device **1400** includes one or more processing units **1402** (e.g., microprocessors, ASICs, FPGAs, GPUs, CPUs, processing cores, and/or the like), one or more input/output (I/O) devices and sensors **1406**, one or more communication interfaces **1408** (e.g., USB, FIREWIRE, THUNDERBOLT, IEEE 802.3x, IEEE 802.11x, IEEE 802.16x, GSM, CDMA, TDMA, GPS, IR, BLUETOOTH, ZIGBEE, SPI, I2C, and/or the like type interface), one or more programming (e.g., I/O) interfaces **1410**, one or more output device(s) **1412**, one or more interior and/or exterior facing image sensor systems **1414**, a memory **1420**, and one or more communication buses **1404** for interconnecting these and various other components.

(72) In some implementations, the one or more communication buses **1404** include circuitry that interconnects and controls communications between system components. In some implementations, the one or more I/O devices and sensors **1406** include at least one of an inertial measurement unit (IMU), an accelerometer, a magnetometer, a gyroscope, a thermometer, one or more physiological sensors (e.g., blood pressure monitor, heart rate monitor, blood oxygen sensor, blood glucose sensor, etc.), one or more microphones, one or more speakers, a haptics engine, one or more depth sensors (e.g., a structured light, a time-of-flight, or the like), and/or the like.

(73) In some implementations, the one or more output device(s) **1412** include one or more displays configured to present a view of a 3D environment to the user. In some implementations, the one or more displays **1412** correspond to holographic, digital light processing (DLP), liquid-crystal display (LCD), liquid-crystal on silicon (LCoS), organic light-emitting field-effect transitory (OLET), organic light-emitting diode (OLED), surface-conduction electron-emitter display (SED), field-emission display (FED), quantum-dot light-emitting diode (QD-LED), micro-electromechanical system (MEMS), and/or the like display types. In some implementations, the one or more displays correspond to diffractive, reflective, polarized, holographic, etc. waveguide displays. In one example, the device **1400** includes a single display. In another example, the device **1400** includes a display for each eye of the user.

(74) In some implementations, the one or more output device(s) **1412** include one or more audio producing devices. In some implementations, the one or more output device(s) **1412** include one or more speakers, surround sound speakers, speaker-arrays, or headphones that are used to produce spatialized sound, e.g., 3D audio effects. Such devices may virtually place sound sources in a 3D environment, including behind, above, or below one or more listeners. Generating spatialized sound may involve transforming sound waves (e.g., using head-related transfer function (HRTF), reverberation, or cancellation techniques) to mimic natural soundwaves (including reflections from walls and floors), which emanate from one or more points in a 3D environment. Spatialized sound may trick the listener's brain into interpreting sounds as if the sounds occurred at the point(s) in the 3D environment (e.g., from one or more particular sound sources) even though the actual sounds may be produced by speakers in other locations. The one or more output device(s) **1412** may additionally or alternatively be configured to generate haptics.

(75) In some implementations, the one or more image sensor systems **1414** are configured to obtain image data that corresponds to at least a portion of a physical environment. For example, the one or more image sensor systems **1414** may include one or more RGB cameras (e.g., with a complimentary metal-oxide-semiconductor (CMOS) image sensor or a charge-coupled device (CCD) image sensor), monochrome cameras, IR cameras, depth cameras, event-based cameras,

and/or the like. In various implementations, the one or more image sensor systems **1414** further include illumination sources that emit light, such as a flash. In various implementations, the one or more image sensor systems **1414** further include an on-camera image signal processor (ISP) configured to execute a plurality of processing operations on the image data.
(76) The memory **1420** includes high-speed random-access memory, such as DRAM, SRAM, DDR RAM, or other random-access solid-state memory devices. In some implementations, the memory **1420** includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid-state storage devices. The memory **1420** optionally includes one or more storage devices remotely located from the one or more processing units **1402**. The memory **1420** comprises a non-transitory computer readable storage medium.
(77) In some implementations, the memory **1420** or the non-transitory computer readable storage medium of the memory **1420** stores an optional operating system **1430** and one or more instruction set(s) **1440**. The operating system **1430** includes procedures for handling various basic system services and for performing hardware dependent tasks. In some implementations, the instruction set(s) **1440** include executable software defined by binary information stored in the form of electrical charge. In some implementations, the instruction set(s) **1440** are software that is executable by the one or more processing units **1402** to carry out one or more of the techniques described herein.
(78) The instruction set(s) **1440** include environment instruction set(s) **1442** configured to, upon execution, identify and/or interpret movements relative to a user interface as described herein. The instruction set(s) **1440** may be embodied as a single software executable or multiple software executables.
(79) Although the instruction set(s) **1440** are shown as residing on a single device, it should be understood that in other implementations, any combination of the elements may be located in separate computing devices. Moreover, the figure is intended more as functional description of the various features which are present in a particular implementation as opposed to a structural schematic of the implementations described herein. As recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated. The actual number of instructions sets and how features are allocated among them may vary from one implementation to another and may depend in part on the particular combination of hardware, software, and/or firmware chosen for a particular implementation.
(80) It will be appreciated that the implementations described above are cited by way of example, and that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope includes both combinations and sub combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.
(81) As described above, one aspect of the present technology is the gathering and use of sensor data that may include user data to improve a user's experience of an electronic device. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies a specific person or can be used to identify interests, traits, or tendencies of a specific person. Such personal information data can include movement data, physiological data, demographic data, location-based data, telephone numbers, email addresses, home addresses, device characteristics of personal devices, or any other personal information.
(82) The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information data can be used to improve the content viewing experience. Accordingly, use of such personal information data may enable calculated control of the electronic device. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure.

(83) The present disclosure further contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information and/or physiological data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. For example, personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection should occur only after receiving the informed consent of the users. Additionally, such entities would take any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices.
(84) Despite the foregoing, the present disclosure also contemplates implementations in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware or software elements can be provided to prevent or block access to such personal information data. For example, in the case of user-tailored content delivery services, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services. In another example, users can select not to provide personal information data for targeted content delivery services. In yet another example, users can select to not provide personal information, but permit the transfer of anonymous information for the purpose of improving the functioning of the device.
(85) Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, content can be selected and delivered to users by inferring preferences or settings based on non-personal information data or a bare minimum amount of personal information, such as the content being requested by the device associated with a user, other non-personal information available to the content delivery services, or publicly available information.
(86) In some embodiments, data is stored using a public/private key system that only allows the owner of the data to decrypt the stored data. In some other implementations, the data may be stored anonymously (e.g., without identifying and/or personal information about the user, such as a legal name, username, time and location data, or the like). In this way, other users, hackers, or third parties cannot determine the identity of the user associated with the stored data. In some implementations, a user may access their stored data from a user device that is different than the one used to upload the stored data. In these instances, the user may be required to provide login credentials to access their stored data.
(87) Numerous specific details are set forth herein to provide a thorough understanding of the claimed subject matter. However, those skilled in the art will understand that the claimed subject matter may be practiced without these specific details. In other instances, methods apparatuses, or systems that would be known by one of ordinary skill have not been described in detail so as not to obscure claimed subject matter.
(88) Unless specifically stated otherwise, it is appreciated that throughout this specification discussions utilizing the terms such as “processing,” “computing,” “calculating,” “determining,” and “identifying” or the like refer to actions or processes of a computing device, such as one or more computers or a similar electronic computing device or devices, that manipulate or transform data represented as physical electronic or magnetic quantities within memories, registers, or other

information storage devices, transmission devices, or display devices of the computing platform.
(89) The system or systems discussed herein are not limited to any particular hardware architecture or configuration. A computing device can include any suitable arrangement of components that provides a result conditioned on one or more inputs. Suitable computing devices include multipurpose microprocessor-based computer systems accessing stored software that programs or configures the computing system from a general-purpose computing apparatus to a specialized computing apparatus implementing one or more implementations of the present subject matter. Any suitable programming, scripting, or other type of language or combinations of languages may be used to implement the teachings contained herein in software to be used in programming or configuring a computing device.
(90) Implementations of the methods disclosed herein may be performed in the operation of such computing devices. The order of the blocks presented in the examples above can be varied for example, blocks can be re-ordered, combined, and/or broken into sub-blocks. Certain blocks or processes can be performed in parallel.
(91) The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or value beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.
(92) It will also be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first node could be termed a second node, and, similarly, a second node could be termed a first node, which changing the meaning of the description, so long as all occurrences of the “first node” are renamed consistently and all occurrences of the “second node” are renamed consistently. The first node and the second node are both nodes, but they are not the same node.
(93) The terminology used herein is for the purpose of describing particular implementations only and is not intended to be limiting of the claims. As used in the description of the implementations and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(94) As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting,” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” may be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.
(95) The foregoing description and summary of the invention are to be understood as being in every respect illustrative and exemplary, but not restrictive, and the scope of the invention disclosed herein is not to be determined only from the detailed description of illustrative implementations but according to the full breadth permitted by patent laws. It is to be understood that the implementations shown and described herein are only illustrative of the principles of the present invention and that various modification may be implemented by those skilled in the art without departing from the scope and spirit of the invention.

Claims

1. A method comprising: at an electronic device having a processor: displaying an extended reality (XR) environment corresponding to a three-dimensional (3D) environment, wherein the XR environment comprises a user interface and a movement, the movement comprising segments; determining an occurrence of an event associated with contact with the user interface in the XR environment; determining whether each of the segments of the movement has a characteristic that satisfies a retraction criterion, the retraction criterion configured to distinguish retraction motion from another type of motion; and associating a subset of the segments of the movement with user interface contact based on whether the characteristic of each of the segments satisfies the retraction criterion.
2. The method of claim 1, wherein the characteristic comprises a retraction confidence determined based on alignment between a direction of the movement during a respective segment and a retraction direction.
3. The method of claim 2, wherein the retraction direction is a direction from a portion of the user to a head of the user.
4. The method of claim 2, wherein the retraction criterion is whether the retraction confidence exceeds a threshold.
5. The method of claim 2, wherein the retraction criterion is whether a change in the retraction confidence exceeds a threshold.
6. The method of claim 1, wherein the retraction criterion comprises whether a portion of the user has stopped moving.
7. The method of claim 1, wherein associating a subset of the segments of the movement with user interface contact comprises generating a retraction dead-band.
8. The method of claim 1, wherein the movement corresponds to a movement of a fingertip or hand.
9. The method of claim 1, wherein the electronic device is a head-mounted device.
10. A system comprising: a non-transitory computer-readable storage medium; and one or more processors coupled to the non-transitory computer-readable storage medium, wherein the non-transitory computer-readable storage medium comprises program instructions that, when executed on the one or more processors, cause the system to perform operations comprising: determining an occurrence of an event associated with contact with the user interface in the XR environment; displaying an extended reality (XR) environment corresponding to a three-dimensional (3D) environment, wherein the XR environment comprises a user interface and a movement, the movement comprising segments; determining whether each of the segments of the movement has a characteristic that satisfies a retraction criterion, the retraction criterion configured to distinguish retraction motion from another type of motion; and associating a subset of the segments of the movement with user interface contact based on whether the characteristic of each of the segments satisfies the retraction criterion.
11. The system of claim 10, wherein the characteristic comprises a retraction confidence determined based on alignment between a direction of the movement during a respective segment and a retraction direction.
12. The system of claim 11, wherein the retraction direction is a direction from a portion of the user to a head of the user.
13. The system of claim 11, wherein the retraction criterion is whether the retraction confidence exceeds a threshold.
14. The system of claim 11, wherein the retraction criterion is whether a change in the retraction confidence exceeds a threshold.
15. The system of claim 10, wherein the retraction criterion comprises whether a portion of the user

has stopped moving.
16. The system of claim 10, wherein associating a subset of the segments of the movement with user interface contact comprises generating a retraction dead-band.
17. The system of claim 10, wherein the movement corresponds to a movement of a fingertip or hand.
18. The system of claim 10, wherein the system is a head-mounted device.
19. A non-transitory computer-readable storage medium storing program instructions executable via one or more processors to perform operations comprising: displaying an extended reality (XR) environment corresponding to a three-dimensional (3D) environment, wherein the XR environment comprises a user interface and a movement, the movement comprising segments; determining an occurrence of an event associated with contact with the user interface in the XR environment; determining whether each of the segments of the movement has a characteristic that satisfies a retraction criterion, the retraction criterion configured to distinguish retraction motion from another type of motion; and associating a subset of the segments of the movement with user interface contact based on whether the characteristic of each of the segments satisfies the retraction criterion.
20. The non-transitory computer-readable storage medium of claim 19, wherein the characteristic comprises a retraction confidence determined based on alignment between a direction of the movement during a respective segment and a retraction direction.
21. The non-transitory computer-readable storage medium of claim 20, wherein the retraction direction is a direction from a portion of the user to a head of the user.
22. The non-transitory computer-readable storage medium of claim 20, wherein the retraction criterion is whether the retraction confidence exceeds a threshold.
23. The non-transitory computer-readable storage medium of claim 20, wherein the retraction criterion is whether a change in the retraction confidence exceeds a threshold.
24. The non-transitory computer-readable storage medium of claim 19, wherein the retraction criterion comprises whether a portion of the user has stopped moving.
25. The non-transitory computer-readable storage medium of claim 19, wherein associating a subset of the segments of the movement with user interface contact comprises generating a retraction dead-band.